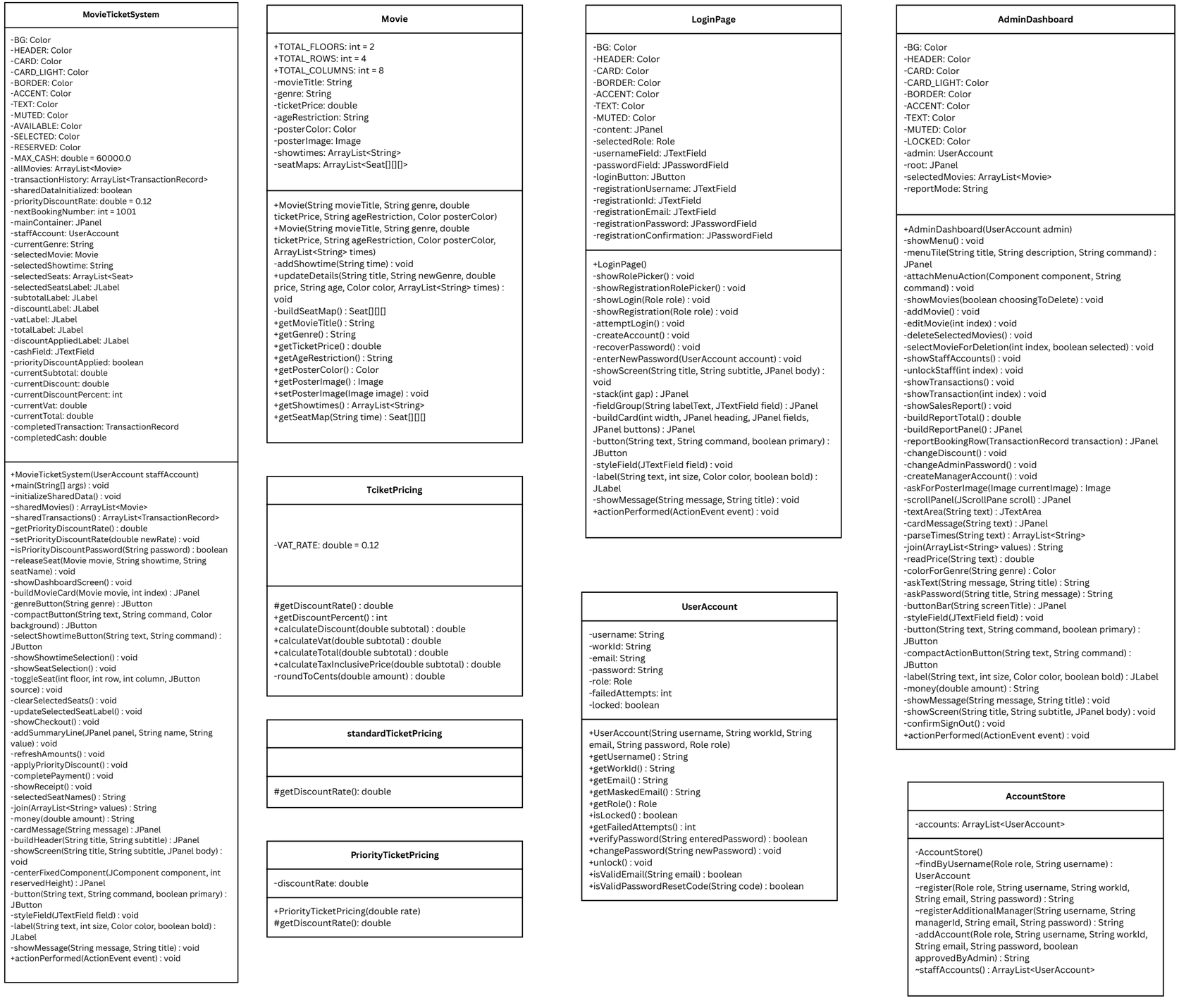

MovieTicketSystem
-BG: Color
-HEADER: Color
-CARD: Color
-CARD_LIGHT: Color
-BORDER: Color
-ACCENT: Color
-TEXT: Color
-MUTED: Color
-AVAILABLE: Color
-SELECTED: Color
-RESERVED: Color
-MAX_CASH: double = 60000.0
-allMovies: ArrayList<Movie>
-transactionHistory: ArrayList<TransactionRecord>
-sharedDataInitialized: boolean
-priorityDiscountRate: double = 0.12
-nextBookingNumber: int = 1001
-mainContainer: JPanel
-staffAccount: UserAccount
-currentGenre: String
-selectedMovie: Movie
-selectedShowtime: String
-selectedSeats: ArrayList<Seat>
-selectedSeatsLabel: JLabel
-subtotalLabel: JLabel
-discountLabel: JLabel
-vatLabel: JLabel
-totalLabel: JLabel
-discountAppliedLabel: JLabel
-cashField: JTextField
-priorityDiscountApplied: boolean
-currentSubtotal: double
-currentDiscount: double
-currentDiscountPercent: int
-currentVat: double
-currentTotal: double
-completedTransaction: TransactionRecord
-completedCash: double
+MovieTicketSystem(UserAccount staffAccount)
+main(String[] args) : void
~initializeSharedData() : void
~sharedMovies() : ArrayList<Movie>
~sharedTransactions() : ArrayList<TransactionRecord>
~getPriorityDiscountRate() : double
~setPriorityDiscountRate(double newRate) : void
~isPriorityDiscountPassword(String password) : boolean
~releaseSeat(Movie movie, String showtime, String seatName) : void
-showDashboardScreen() : void
-buildMovieCard(Movie movie, int index) : JPanel
-genreButton(String genre) : JButton
-compactButton(String text, String command, Color background) : JButton
-selectShowtimeButton(String text, String command) : JButton
-showShowtimeSelection() : void
-showSeatSelection() : void
-toggleSeat(int floor, int row, int column, JButton source) : void
-clearSelectedSeats() : void
-updateSelectedSeatLabel() : void
-showCheckout() : void
-addSummaryLine(JPanel panel, String name, String value) : void
-refreshAmounts() : void
-applyPriorityDiscount() : void
-completePayment() : void
-showReceipt() : void
-selectedSeatNames() : String
-join(ArrayList<String> values) : String
-money(double amount) : String
-cardMessage(String message) : JPanel
-buildHeader(String title, String subtitle) : JPanel
-showScreen(String title, String subtitle, JPanel body) : void
-centerFixedComponent(JComponent component, int reservedHeight) : JPanel
-button(String text, String command, boolean primary) : JButton
-styleField(JTextField field) : void
-label(String text, int size, Color color, boolean bold) : JLabel
-showMessage(String message, String title) : void
+actionPerformed(ActionEvent event) : void
Movie
+TOTAL_FLOORS: int = 2
+TOTAL_ROWS: int = 4
+TOTAL_COLUMNS: int = 8
-movieTitle: String
-genre: String
-ticketPrice: double
-ageRestriction: String
-posterColor: Color
-posterImage: Image
-showtimes: ArrayList<String>
-seatMaps: ArrayList<Seat[][]>
+Movie(String movieTitle, String genre, double ticketPrice, String ageRestriction, Color posterColor)
+Movie(String movieTitle, String genre, double ticketPrice, String ageRestriction, Color posterColor, ArrayList<String> times)
-addShowtime(String time) : void
+updateDetails(String title, String newGenre, double price, String age, Color color, ArrayList<String> times) : void
-buildSeatMap() : Seat[][][]
+getMovieTitle() : String
+getGenre() : String
+getTicketPrice() : double
+getAgeRestriction() : String
+getPosterColor() : Color
+getPosterImage() : Image
+setPosterImage(Image image) : void
+getShowtimes() : ArrayList<String>
+getSeatMap(String time) : Seat[][][]
TciketPricing
-VAT_RATE: double = 0.12
#getDiscountRate() : double
+getDiscountPercent() : int
+calculateDiscount(double subtotal) : double
+calculateVat(double subtotal) : double
+calculateTotal(double subtotal) : double
+calculateTaxInclusivePrice(double subtotal) : double
-roundToCents(double amount) : double
standardTicketPricing
#getDiscountRate(): double
PriorityTicketPricing
-discountRate: double
+PriorityTicketPricing(double rate)
#getDiscountRate(): double
LoginPage
-BG: Color
-HEADER: Color
-CARD: Color
-BORDER: Color
-ACCENT: Color
-TEXT: Color
-MUTED: Color
-content: JPanel
-selectedRole: Role
-usernameField: JTextField
-passwordField: JPasswordField
-loginButton: JButton
-registrationUsername: JTextField
-registrationId: JTextField
-registrationEmail: JTextField
-registrationPassword: JPasswordField
-registrationConfirmation: JPasswordField
+LoginPage()
-showRolePicker() : void
-showRegistrationRolePicker() : void
-showLogin(Role role) : void
-showRegistration(Role role) : void
-attemptLogin() : void
-createAccount() : void
-recoverPassword() : void
-enterNewPassword(UserAccount account) : void
-showScreen(String title, String subtitle, JPanel body) : void
-stack(int gap) : JPanel
-fieldGroup(String labelText, JTextField field) : JPanel
-buildCard(int width, JPanel heading, JPanel fields, JPanel buttons) : JPanel
-button(String text, String command, boolean primary) : JButton
-styleField(JTextField field) : void
-label(String text, int size, Color color, boolean bold) : JLabel
-showMessage(String message, String title) : void
+actionPerformed(ActionEvent event) : void
UserAccount
-username: String
-workId: String
-email: String
-password: String
-role: Role
-failedAttempts: int
-locked: boolean
+UserAccount(String username, String workId, String email, String password, Role role)
+getUsername() : String
+getWorkId() : String
+getEmail() : String
+getMaskedEmail() : String
+getRole() : Role
+isLocked() : boolean
+getFailedAttempts() : int
+verifyPassword(String enteredPassword) : boolean
+changePassword(String newPassword) : void
+unlock() : void
+isValidEmail(String email) : boolean
+isValidPasswordResetCode(String code) : boolean
AdminDashboard
-BG: Color
-HEADER: Color
-CARD: Color
-CARD_LIGHT: Color
-BORDER: Color
-ACCENT: Color
-TEXT: Color
-MUTED: Color
-LOCKED: Color
-admin: UserAccount
-root: JPanel
-selectedMovies: ArrayList<Movie>
-reportMode: String
+AdminDashboard(UserAccount admin)
-showMenu() : void
-menuTile(String title, String description, String command) : JPanel
-attachMenuAction(Component component, String command) : void
-showMovies(boolean choosingToDelete) : void
-addMovie() : void
-editMovie(int index) : void
-deleteSelectedMovies() : void
-selectMovieForDeletion(int index, boolean selected) : void
-showStaffAccounts() : void
-unlockStaff(int index) : void
-showTransactions() : void
-showTransaction(int index) : void
-showSalesReport() : void
-buildReportTotal() : double
-buildReportPanel() : JPanel
-reportBookingRow(TransactionRecord transaction) : JPanel
-changeDiscount() : void
-changeAdminPassword() : void
-createManagerAccount() : void
-askForPosterImage(Image currentImage) : Image
-scrollPanel(JScrollPane scroll) : JPanel
-textArea(String text) : JTextArea
-cardMessage(String text) : JPanel
-parseTimes(String text) : ArrayList<String>
-join(ArrayList<String> values) : String
-readPrice(String text) : double
-colorForGenre(String genre) : Color
-askText(String message, String title) : String
-askPassword(String title, String message) : String
-buttonBar(String screenTitle) : JPanel
-styleField(JTextField field) : void
-button(String text, String command, boolean primary) : JButton
-compactActionButton(String text, String command) : JButton
-label(String text, int size, Color color, boolean bold) : JLabel
-money(double amount) : String
-showMessage(String message, String title) : void
-showScreen(String title, String subtitle, JPanel body) : void
-confirmSignOut() : void
+actionPerformed(ActionEvent event) : void
AccountStore
-accounts: ArrayList<UserAccount>
-AccountStore()
~findByUsername(Role role, String username) : UserAccount
~register(Role role, String username, String workId, String email, String password) : String
~registerAdditionalManager(String username, String managerId, String email, String password) : String
-addAccount(Role role, String username, String workId, String email, String password, boolean approvedByAdmin) : String
~staffAccounts() : ArrayList<UserAccount>